

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech.)

Semester - II Examination May 2010

B.Tech. (CE/EE/IT) Regular Exam &

B.Tech. (CL/EC/ME) Backlog Examination

CL - 101 Fundamentals of Civil Engineering

Date: 13.05.2010, Thursday

Time: 12:30 p.m. To 03:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Write the uses of surveying. Differentiate between prismatic compass and surveyor's compass. [5]

(b) Define: (i) Magnetic declination (ii) True meridian [2]

Q - 2 (a) The following staff readings were taken with levelling instruments at an interval of 20 m. The readings are 2.385, 1.740, 0.625, 3.550, 2.745, 2.075, 1.845, 0.975, 0.545, 1.640, 2.355 and 3.740. The instrument was shifted after 4th and 8th reading. The last reading was taken on a B.M. of R.L. 120.250 m. Find the R.L.s of all the points by Rise and Fall method. Also apply arithmetic checks. [5]

(b) Following are the bearings of lines of a closed traverse ABCD. [5]

Line	F. B.	B. B.
AB	N 45° 30' E	S 45° 30' W
BC	S 60° 00' E	N 60° 00' W
CD	S 10° 30' W	N 10° 30' E
DA	N 75° 45' W	S 75° 45' E

Calculate the interior angles of the traverse and apply the necessary checks.

(c) (i) Differentiate between Whole Circle Bearing and Reduced Bearing. [2]

(ii) Convert given W.C.B. to R.B. and R.B. to W.C.B. [2]

(i) 60° (ii) 240° (iii) S 47° E (iv) N 25° W

OR

Q - 2 (a) (i) Write a short note on Geographic Information System. [2]

(ii) A 30.00 m tape is suspended between the ends under a pull of 15 kg. The weight of the tape is 1.8 kg. Find the corrected length of the tape between its ends. [3]

- (b) The following readings were obtained, when the area was measured by planimeter. [3]

The tracing arm being set to 100 cm^2 as the value of one revolution of roller.

I.R.	F.R.	Position of Anchor Arm	Remark
7.285	3.225	Outside the figure	The zero of the disc passed the fixed index mark once in clock wise direction
1.250	4.370	Inside the figure	The zero of the disc passed the fixed index mark twice in reverse direction

- (c) The following bearings were observed while traversing with compass: [6]

Line	F. B.	B. B.
AB	$59^\circ 00'$	$239^\circ 00'$
BC	$139^\circ 30'$	$317^\circ 00'$
CD	$215^\circ 15'$	$36^\circ 30'$
DE	$208^\circ 00'$	$29^\circ 00'$
EA	$318^\circ 30'$	$138^\circ 45'$

Find which stations are affected by local attraction? Determine the corrected bearings.

- Q - 3 (a) Define: Traffic Engineering. What is the object of traffic engineering? Discuss the factors affecting traffic. [4]

- (b) Draw a neat sketch of permanent way (railway track). What is Gauge? Enlist types of gauges with dimension. [5]

- (c) A line was measured with a steel tape, which was exactly 30.00 m at a pull of 5 kg and the measured length was 230.00 m. The pull applied during measurement was 12 kg and the tape was uniformly supported. Find the true length of the line, if the cross-sectional area of the tape was 0.02 cm^2 . The Modulus of Elasticity = $2.1 \times 10^6 \text{ kg / cm}^2$. [5]

OR

- Q - 3 (a) How the transportation helps in development of the country? Discuss in detail. [3]

- (b) What are the advantages and disadvantages of roadways, railways and airways? [6]

- (c) A survey line CDE crosses the river, D being on the near river bank and E on the opposite side. A perpendicular, DF = 100 m is ranged at D on the left. From F, bearings of E and C are observed to be 40° and 130° respectively. Chainage of C is 5000 m and that of D is 5050 m. Find the chainage of E. [5]

SECTION – II

- Q - 4 (a)** Draw the detailed **plan** of a room having a size of 5 m × 4 m showing proper locations and sizes of doors, windows and ventilators. Take wall thickness = 30 cm. [4]
- (b)** List out various branches of civil engineering. Discuss the importance of each branch. [3]
- Q - 5 (a)** Which are the applications of hydrology in real field? [3]
- (b)** Attempt any **Two** short notes: [6]
- (i) Canal regulation works
 - (ii) Basic requirements of building planning
 - (iii) Design loads acting on a structure
- (c)** What do you mean by conservation of water? Discuss water conservation measures for domestic, industrial and irrigation conservation. [5]

OR

- Q - 5 (a)** Classify the canals by different ways. [3]
- (b)** Define: Evaporation, Transpiration, Infiltration, Aquifer, Aquitard [5]
- (c)** Write short notes on: [6]
- (i) Building bye-laws
 - (ii) B.O.T. projects
 - (iii) Sub-surface water sources
- Q - 6 (a)** Draw a layout for a chemical industry. [5]
- (b)** Discuss scope of civil engineering. [3]
- (c)** Explain the following principles of building planning briefly: [6]
- (i) Aspect (ii) Flexibility (iii) Grouping (iv) Prospect (v) Roominess (vi) Sanitation

OR

- Q - 6 (a)** Draw a layout for a textile industry. [5]
- (b)** Differentiate load bearing structure and frame structure. [3]
- (c)** Enlist all the building components. Write the functions of any five building components. [6]
